

Topics to discuss

- Weddle's Rule
- Numerical Problem
- Homework Problem

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Weddle's Rule :

Weddle's rule is a high-order Newton-Cotes method used to approximate a definite integral by fitting a polynomial through 7 equally spaced points (i.e. 6 sub-intervals) and integrating that polynomial.

$$h = \frac{b-a}{n}, \quad n \text{ must be multiple of } 6.$$

The formula is,

$$\int_{x_0}^{x_6} f(x) dx = \frac{3h}{10} [y_0 + y_6 + 5(y_1 + y_5) + (y_2 + y_4) + 6y_3]$$

Numerical Problem :

Q: Use Weddle's rule to find approximate; $\int_0^6 x^2 dx$
Take $n=6$.

Solution. Let, $y = f(x) = x^2$

$$\text{Step length, } h = \frac{b-a}{n} = \frac{6-0}{6} = 1$$

$$\text{So, } x_0 = 0$$

$$x_1 = x_0 + h = 1$$

$$x_2 = x_0 + 2h = 2$$

$$\vdots$$

$$x_6 = x_0 + 6h = 6$$

We have, $f(x) = x^2$

x	0	1	2	3	4	5	6
$f(x)$	0	1	4	9	16	25	36

Applying Weddle's rule,

$$\begin{aligned}\int_0^6 x^2 dx &= \frac{3h}{10} [y_0 + y_6 + 5(y_1 + y_5) + (y_2 + y_4) + 6y_3] \\ &= \frac{3 \times 1}{10} [0 + 36 + 5(1 + 25) + (4 + 16) + 6 \times 9] \\ &= 72 \text{ Ans}\end{aligned}$$

Homework Problem

Q: Use Weddle's rule to approximate: $\int_0^6 \frac{1}{1+x} dx$.

Solution:-

$$\text{Let, } y = f(x) = \frac{1}{1+x}$$

$$\text{Step length, } h = \frac{b-a}{n} = \frac{6-0}{6} = 1$$

As it's Weddle's rule, so we are taking $n=6$

We have, $f(x) = \frac{1}{1+x}$

x	0	1	2	3	4	5	6
$f(x)$	1	0.5	0.3333	0.25	0.200	0.1667	0.1429

Applying Weddle's rule,

$$\begin{aligned}
 \int_1^6 f(x) dx &= \frac{3h}{10} [y_0 + y_6 + 5(y_1 + y_5) + (y_2 + y_4) + 6y_3] \\
 &= \frac{3 \times 1}{10} [1 + 0.1429 + 5(0.5 + 0.1667) + (0.333 + 0.200) + 6 \times 0.25] \\
 &= 1.952910
 \end{aligned}$$

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